

Attention Is All You Need

The transformer Architecture

Google Brain Team 2017

What's the Paper About?

- Introduces a completely new neural network architecture called the **Transformer**
- Old models (RNN, LSTM) were like reading a sentence one word at a time, slowly
- The Transformer reads ALL words at the same time, much faster
- It uses "attention" - a technique where the AI focuses on important words

Why We Needed Something New

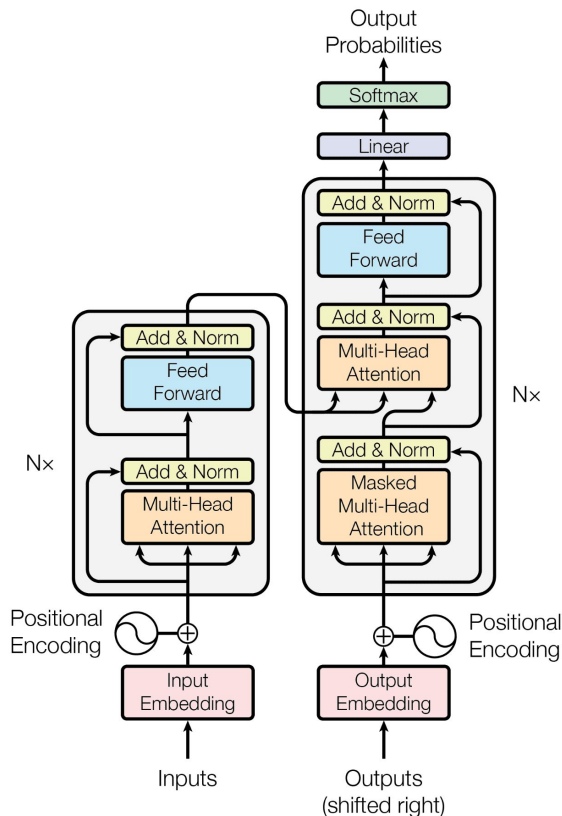
- **RNNs are slow:** They process word-by-word (can't do them in parallel)
- **Memory problems:** When sentences are long, they forget what they read earlier
- **Expensive training:** Takes days on expensive GPUs
- **Long-range issues:** Can't connect words that are far apart well

What is Attention?

- Think of it like reading: You don't focus equally on every word. You focus on what matters.
- Example: In "The cat sat on the mat," you might focus on "cat" and "sat" more than "the"
- In AI: Each word learns which other words are important to pay attention to

The Transformer Architecture

- **Encoder:** Takes input (like English text) and understands it using attention



- **Decoder:** Creates output (like German text) while looking at what the encoder understood

Self-Attention

- For each word, the model asks: "Which other words should I focus on?"
- **Query** (Q): "What am I looking for?"
- **Key** (K): "What does each word offer?"
- **Value** (V): "What information should I actually take from each word?"

How Attention Mathematically Works

- The formula $\text{Attention}(Q, K, V) = \text{softmax}(Q \times K^T / \sqrt{d}) \times V$
 - a. Compare Query with all Keys (find relevance scores)
 - b. Normalize scores so they add to 1 (softmax)
 - c. Use scores as weights to blend Values
- Example: If word A has score 0.7 and word B has score 0.3, take 70% from A and 30% from B

Multi-Head Attention

- Instead of one "attention lens," use 8 different lenses simultaneously (the "heads")
- Different heads learn different things:
 - a. Head 1 might focus on grammar relationships
 - b. Head 2 might focus on semantic meaning
 - c. Head 3 might focus on pronouns and their references
 - d. ...etc
- Why it works: Different perspectives give richer understanding
- Combine all 8 perspectives together for the final answer

Stacked Layers

- Both encoder and decoder have 6 identical layers stacked on top of each other
- Each layer has:
 - a. Attention: Multi-head attention (8 heads)
 - b. Feed-forward: A simple neural network that processes each word
- Residual connections: Skip connections that help information flow better (like shortcuts in a building)
- Layer normalization: Keeps the numbers stable during training

Positional Encoding

- **Problem:** Without RNNs, the model doesn't know word order.
- **Solution:** Add position information using sine/cosine functions.
- **Example:** "The cat sat" vs "sat cat The" - positional encoding makes them different

Advantages Over Old Models

- **Parallelizable:** Process all words at once, not one-by-one
- **Constant path length:** Every word can "see" every other word instantly
- **Faster training:** Trained in 12 hours vs days for RNNs
- **No forgetting:** Each word can access any earlier word, so no "forgetting" problem

Results

- **English → German translation:** Scored 28.4 (2 points better than previous best)
- **English → French translation:** Scored 41.8 (new record!)
- These are huge improvements in machine translation quality

Training Details

- **Hardware:** 8 NVIDIA P100 GPUs
- **Base model:** Trained in 12 hours
- **Big model:** Trained in 3.5 days
- **Optimizer:** Adam with a custom learning rate schedule
- Added dropout (randomly ignore some neurons) to prevent overfitting

Beyond Translation

- The paper showed Transformers aren't just for translation
- Tested on parsing (understanding sentence structure)
- Got competitive results, showing it's a general-purpose architecture

Why This Paper Changed Everything

- **Foundation for GPT:** Chat GPT is built on Transformers
- **Foundation for BERT:** Used in millions of applications
- **Simpler design:** Actually easier to understand than RNNs
- **Scales beautifully:** Bigger models = better results (this enabled modern LLMs)
- **Sparked research:** Thousands of papers built on this

Thank You !